

ANSI/IES LM-9-20(R2023)



Illuminating
ENGINEERING SOCIETY

APPROVED METHOD:
ELECTRICAL AND PHOTOMETRIC
MEASUREMENT OF
FLUORESCENT LAMPS
AN AMERICAN NATIONAL STANDARD



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Publication of this Approved Method
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared by:
The IES Testing Procedures Committee**



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Foreword

This Approved Method is an update to *IES LM-9-09(R17)*, *IESNA Approved Method for the Electrical and Photometric Measurement of Fluorescent Lamps*. In addition to linear fluorescent lamps, this document covers all U-bent and circular fluorescent lamps. Guidance is provided regarding test requirements and to promote uniformity in measurement procedures.

1.0 Introduction and Scope

1.1 Introduction

The fluorescent lamp is an electric discharge source in which light is produced predominantly by fluorescent powders activated by ultraviolet energy generated by a mercury arc. Like most electric discharge lamps, fluorescent lamps must be operated either in series with a current limiting device or from an electronic circuit. This device, commonly called electromagnetic or electronic ballast, limits the arc current to the value for which each lamp type is designed. The ballast in conjunction with the appropriate power source provides the required starting and operating lamp voltages. These methods are applicable to lamps having hot cathodes and to lamps of the cold cathode variety.

For special purposes, it may be desirable to determine the characteristics of lamps when they are operated at other than the standard conditions described in this Approved Method. Where this is done, such results are meaningful only for the particular conditions under which they were obtained and these conditions shall be stated in the test report.

The photometric information usually required is total luminous flux (lumens), luminous intensity (candelas) in one or more directions, and color. For the purpose of this Approved Method, the determination of these data will be considered photometric measurements.

The electrical characteristics usually measured are line voltage, lamp voltage, lamp current, and lamp power. In the case of rapid start lamps, the power measurements may include both the arc watts and the cathode power. Arc watts is the power consumed by the lamp discharge only and is exclusive of any power that may be supplied to the lamp cathodes from a separate voltage source. Total lamp power will be the sum of arc watts and cathode watts. For the purpose of this Approved Method, the determination of these data will be considered electrical measurements.

1.2 Scope

This Approved Method describes the procedures to be followed and the precautions to be observed in obtaining uniform and reproducible measurements of the electrical and photometric characteristics of fluorescent lamps under standard conditions in 60 Hz, alternating current circuits and under high frequency conditions (reference high frequency circuits are operated at 25 kHz). Single-ended compact fluorescent lamps are covered in LM-66¹ and are excluded from this procedure.

2.0 Normative References

The most current versions of:

- American National Standards Institute, *Specification for Fluorescent Lamp Reference Ballasts*, ANSI C82.3. New York, NY: ANSI.
- American National Standards Institute, *Standard Guide for Electrical Measurements of Fluorescent Lamps*, ANSI C78.375. New York, NY: ANSI.
- American National Standards Institute, *Standard Guide for Electric Lamps, Double-Capped Fluorescent Lamps - Dimensional and Electrical Characteristics*, ANSI C78.81. New York, NY: ANSI.
- Subcommittee on Photometry of Light Sources of the IES testing Procedures Committee, *IES Guide to Lamp Seasoning*, LM-54. New York, NY: Illuminating Engineering Society.

3.0 Nomenclature and Definitions

The units of electrical measurement used in this approved method are the *volt*, the *ampere*, and the *watt*. The units of photometric measurement are the *lumen* and the *candela* (see ANSI/IES LS-1-20, *Nomenclature and Definitions for Illuminating Engineering*¹⁸). Color is specified in terms of CIE recommended systems.^{2,3} The industry-accepted point for making an initial rating of a discharge lamp occurs after the lamp's first 100 hours of operation (ANSI C78.81).

The *cold chamber* or *cold spot* is the location inside a fluorescent lamp with the lowest operating temperature. Both the location and temperature of the cold spot are critical for optimizing lamp performance.

4.0 Ambient and Physical Conditions

4.1 General

Photometric values and electrical characteristics of fluorescent lamps are sensitive to ambient temperature and air movement.

4.2 Ambient Temperature

The ambient temperature shall be measured in the sphere at a point not more than 1 m from the lamp and at the same height as the lamp. Under standard conditions this temperature shall be maintained at $25\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F} \pm 1.8\text{ }^{\circ}\text{F}$). Also, the temperature sensor shall be shielded from the lamp and radiation from any other source. Measurements at temperatures other than $25\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F}$) are considered non-standard and shall be so noted in the test report.

4.3 Air Movement

Air movement at the surface of a lamp under test will substantially alter electrical and photometric values. No discernible airflow from sources other than the device under test shall be allowed. This can be tested using a single ply tissue paper hanging in place of the lamp. To assure stabilized and uniform air convection

over the lamp surface, the lamp should be operated in a horizontal orientation when measurements are being taken. Lamp manufacturers may specify a certain horizontal orientation, which, according to their specifications, would help stabilize the lamp and assure maximum light output. This is particularly true of lamps having a non-circular cross-section.

5.0 Electrical Conditions

5.1 Waveshape

The AC power source, while operating the test lamp, shall have a sinusoidal voltage waveshape such that the RMS summation of the harmonic components does not exceed 3.0 percent of the fundamental.

5.2 Voltage Regulation

The RMS voltage applied to the test lamp for the duration of the measurement interval shall be regulated to within ± 0.1 percent for line voltage measurements and ± 1.0 percent for high frequency measurements.

5.3 Power Source Impedance

The power source impedance for low frequency supplies, measured at the points where the ballast and lamp are connected, shall not exceed 2.0 percent of the ballast impedance. For high frequency power supplies it is not possible to meet the 2.0 percent limit internally in the power supply so electronic control circuits are used to keep the output voltage at the desired level. To ensure compliance with this requirement, variable autotransformers or other voltage transforming devices shall have a kVA rating of at least five times the lamp wattage.

5.4 Ballast

When the electrical or photometric characteristics of a non-integrated fluorescent lamp are being measured for rating purposes, the lamp shall be operating in conjunction with a reference ballast at the specified source voltage. The reference ballast may be a fixed impedance ballast with the proper impedance and power factor for the lamp under test or it may be a variable ballast where resistance and inductance* can be separately adjusted to provide the proper total

impedance and power factor for the lamp under test. The general characteristics of the reference ballasts are described in ANSI C82.3. The reference ballast values for voltage and impedance are given on the ANSI C78.81 series lamp data sheets. If readings are to be taken on a lamp type for which no standard exists, the ballast used shall comply with the general requirements of ANSI C82.3.

For electrical and photometric measurements of fluorescent lamps specified for high frequency operation it is necessary to operate the lamps in series with a non-inductive reference resistor ballast. (See relevant ANSI C78.81 series lamp data sheets for specification of the resistive impedance.) When no resistive impedance is specified the resistive value should be set to one half of the lamp impedance under high frequency conditions.

It may be desirable to determine the characteristics of lamps when operated from ballasts other than reference ballasts. Where this is done, such results are meaningful only for the particular ballast and circuit with which they were obtained. Such results are not directly comparable with data taken on a reference ballast. Non-reference ballasts shall be documented in the test report.

5.5 Circuits and Lamp Connections

5.5.1 Measurement Circuit Connections. As shown in **Figure 1a**, a test instrument capable of measuring and displaying wattage, current, and voltage can be connected to the circuit. The potential element of the instrument is connected on the lamp side of the current measuring element. Switch S_6 provides a means of measuring either lamp or input voltage of the reference circuit (E_1) with the same potential element. An auxiliary voltmeter, V_2 , can be used if monitoring input (E_1) and lamp voltage simultaneously is desired. Corrections to

compensate for the presence of the measuring elements in the circuit can be calculated from the impedance specification data provided by the manufacturer of the instrument, or they can be determined by using switches to remove instruments from the circuit. Usually this type of multifunction instrument has a negligible effect on the circuit and does not require corrections.

5.5.2 Preheat Start Lamps (Line Frequency). For preheat start lamps, the complete reference ballast test circuit is a combination of **Figure 1a** connected (at points "X" and "Y") to **Figure 1b**. If the starting switch S_7 is normally shunted by a capacitor when the lamp is started, this capacitor shall be disconnected before any readings of lamp characteristics are taken.

Lamps of the preheat start type are provided with two contact pins at each end of the lamp. One pin at each end is connected to the operating circuit (at points "X" and "Y" – see **Figure 1a**) and the remaining two pins are connected to the starter circuit switch S_7 (see **Figure 1b**). For consistent results, each lamp shall always be operated with the same two pins connected to the operating circuit. This includes not only the operation of the lamp while measurements are being taken, but also all other periods of operation prior to the time of measurement (e.g., during life testing or on warm-up racks). One pin at each end of the lamp shall be marked to indicate the operating circuit connection point, using the crossover scheme, shown in **Figure 1b**.

5.5.3 Preheat Start Lamps (High Frequency). High frequency operated lamps are four pin lamps without an internal starting circuit. Starting is performed by electronic circuit. The lamps shall always be operated with the pins connected to the same points in the operating circuit as described in **Section 5.5.2** (see circuit diagram in **Figure 2**). Wires shall be of low loss, low capacitance types and shall be kept as short as possible to avoid the effect of the parasitic capacitance or keeping them at low level. Using any unnecessary elements, like optional switches, relays or additional equipment etc. shall be avoided and they shall be omitted from the measuring circuit, particularly from the lamp side of the high frequency reference ballast.

* Variable linear reactors are available from:

NWL Transformers	Warner Power
312 Rising Sun Road	40 Depot Street
Bordentown, NJ 08505	Warner, NH 03278
www.nwl.com	www.warnerpower.com

Note: The sources for linear reactors given here are known to the authors of this Approved Method as of March 2008. Providing this information does not in any way imply an endorsement of these companies. Linear reactors suitable for use as reference ballasts may be available from other sources.

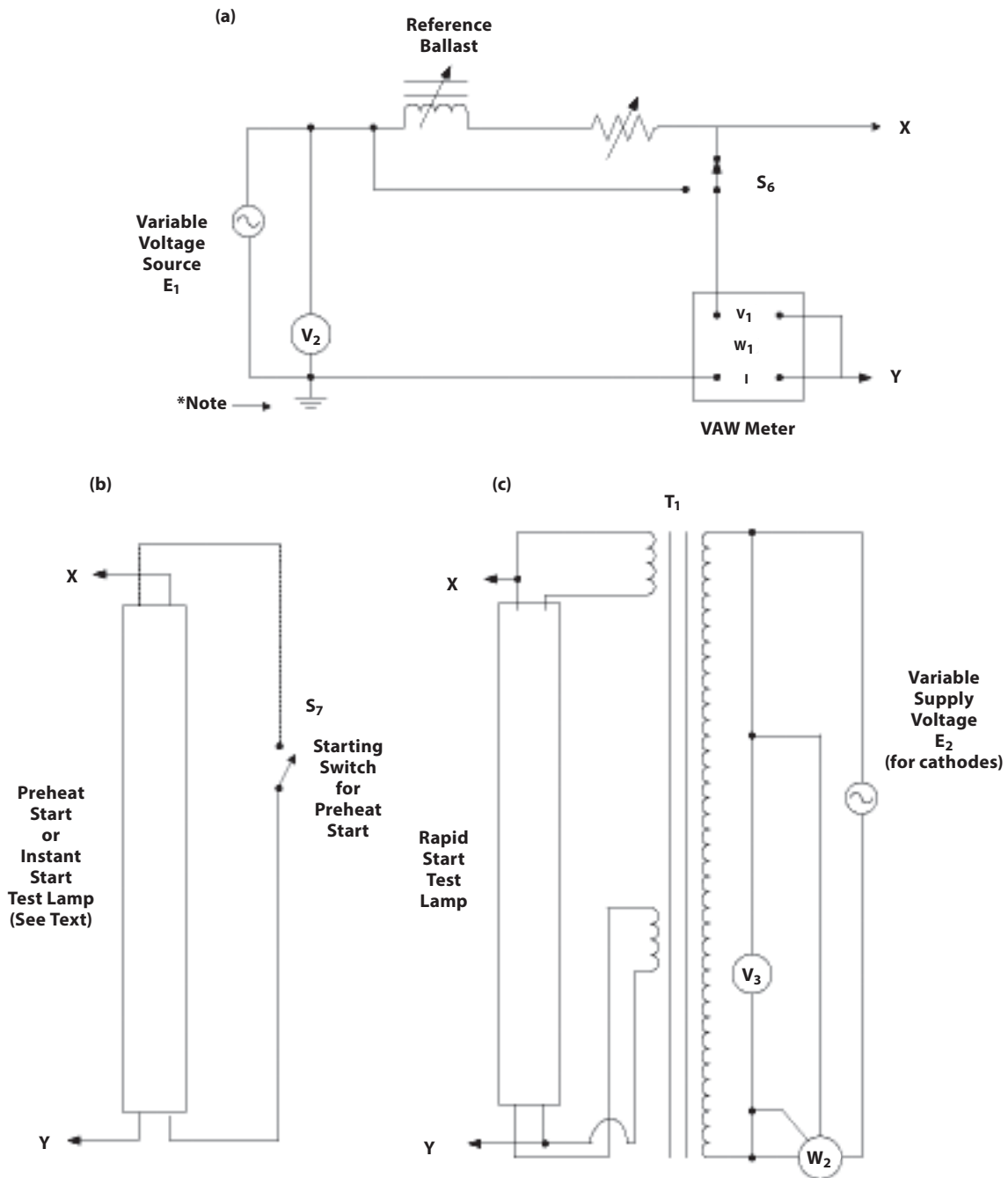


Figure 1. The basic measurement circuit for fluorescent lamps (a) uses a variable voltage source (E_1) in series with a reference ballast. The setup uses a multipurpose VAW meter. The test lamp connects to the measurement circuit at points "X" and "Y". Connections for a preheat start or instant start test lamp (b) show that the preheat start requires starting switch S_7 , while an instant start test lamp does not. Connections for rapid start test lamp (c) include instrumentation for determining the cathode power consumed. *Note: Ground the basic measurement circuit where indicated if a grounded supply system is used.

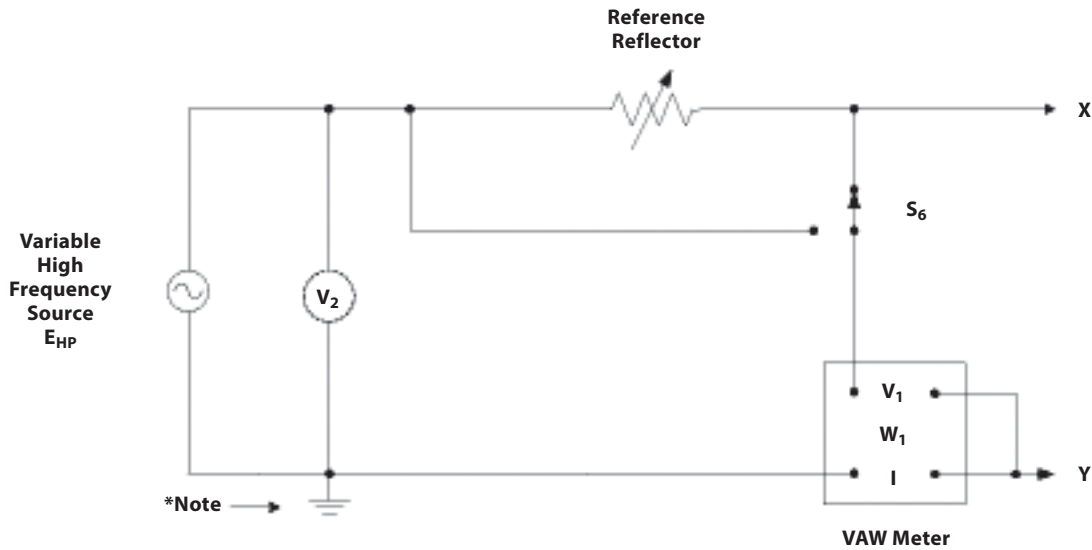


Figure 2. The high frequency measurement circuit for fluorescent lamps uses a variable high frequency voltage source (EHF) in series with reference resistor. The setup uses a multipurpose VAW meter. The test lamp connects to the circuit at points "X" and "Y". *Note: Ground the high frequency measurement circuit where indicated if a grounded supply system is used.

5.5.4 Instant Start Lamps. For instant start lamps, the complete reference ballast test circuit is a combination of **Figure 1a** connected (at points "X" and "Y") to **Figure 1b** (with starting switch S_7 deleted).

5.5.5 Rapid Start Lamps. For rapid start lamps, the complete reference ballast test circuit is a combination of **Figure 1a** connected (at points "X" and "Y") to **Figure 1c**. This way, a separate low voltage cathode power source is available, allowing the lamp to operate normally. The cathode voltage may be supplied from a suitable transformer, T_1 (or from two separate transformers) having known losses. The windings of T_1 shall be connected so that all cathode transformer voltages subtract from the voltage supplied by the reference ballast circuit. Thus the points "X" and "Y" in **Figure 1c** will be the points of maximum potential difference at the lamp holder terminals, and the arc will operate between these two points.

Rapid start lamps shall have the same marking of pins as was described in **Section 5.5.2** for the preheat start lamps, and the marked pins shall always be connected to the same point in the operating circuit. For a reference ballast circuit, this will be the points marked "X" and "Y" in **Figure 1a** and **Figure 1c**. For a commercial ballast circuit, where each lamp is connected to four of the leads

from the ballast, the marked pins will be connected to those two leads having the highest voltage. Multi-lamp ballasts require lamps (or suitable loads) be connected in order to ascertain the points of largest voltage difference for each lamp circuit.

5.5.6 Cathode Power Measurements. The power consumed by the cathodes in an operating rapid start lamp is that indicated by wattmeter W_2 less the transformer loss. The power consumed by the lamp discharge in an operating rapid start lamp is that indicated by wattmeter W_1 and is usually referred to as arc watts. The total lamp power is the sum of W_1 and W_2 less the transformer loss:

$$\text{Total rapid start lamp watts} = (\text{arc watts}) + (\text{cathode watts}) - (\text{transformer loss in watts})$$

The cathode transformer(s) in **Figure 1c** (represented by T_1) shall be calibrated to determine their power loss under normal load condition. This calibration is accomplished by putting dummy load resistors of the proper value, (see the appropriate ANSI specifications^{4,5} for the particular lamp under test), across the secondary windings of T_1 and then adjusting the primary voltage until the average of the two secondary voltages equals the ANSI-specified value. Alternatively, separate cathode transformers may

be used, each with its own adjustable source, permitting individual voltage adjustment for each cathode. However, when separate transformers are used, there are issues associated with harmonizing the phase of the separate power supplies that shall be considered.

Obtain cathode supply voltage (E_2) from the same line (and same phase if a three-phase supply is used) as the reference ballast supply voltage (E_1). Unless this is precaution is observed, the subtractive polarity of the cathode windings may not be obtained.

In **Figure 1c** the input voltage of T_1 is read on V_3 and the input power of T_1 on W_2 . The transformer loss is calculated by subtracting from the input power the losses in the meters and the power absorbed by the dummy load resistors (E^2/R for each of the resistors).

To determine the lamp cathode power, the lamp is operated normally on the reference ballast with the lamp connections shown in **Figure 1**. Primary voltage V_3 is adjusted (as was reported in the previous paragraphs) to give the proper voltage on the cathodes, and the input power is read on W_2 . Cathode power is then equal to the input power minus the losses (as was described in the previous paragraphs).

For high frequency circuits cathode heat is not utilized

6.0 Lamp Test Procedures

6.1 Lamp Orientation

Lamp seasoning, pre-burning, and photometric measurements, with the exception of T5 lamps (see **Section 6.2.2**), shall be done in the same orientation. The standard orientation is the horizontal position unless otherwise specified.

6.2 Lamp Stabilization

Prior to any photometric testing, fluorescent lamps shall be properly seasoned and pre-burned. They shall also be stabilized in the test position. Equilibrium may need to be re-established should a warmed-up lamp be switched to a different ballast.

6.2.1 Seasoning. Prior to taking initial measurements, all new fluorescent lamps selected for test shall be seasoned per *ANSI/IES LM-54-20, Approved Method: Guide to Lamp Seasoning*.

6.2.2 Pre-burning. Before any measurements are taken, lamps shall be operated long enough to reach stabilization and temperature equilibrium. Stability depends not only on the warm-up time but also on the type of source being tested, its operating history, and the individual characteristics of the source. A period of 15 minutes continuous operation is usually sufficient. Many lamp types designed with controlled cold spots, including those which run at 800 mA or more, may require an operating period of several hours,⁶ particularly when the lamps are new or have been handled so as to dislodge excess mercury from the cool zones. However, it is always better to judge stability from periodic checks of light output, lamp voltage or both rather than elapsed time. When light output, and/or lamp voltage become stable, the lamp is stable.

T5 linear fluorescent lamps are designed with cold spot control at the monogrammed end of each lamp. All manufactures adhere to this convention. The lumen output of the T5 lamps maximizes in ambient still air at 35 °C (95 °F). In order to obtain stable photometric results in 25 °C (77 °F) ambient air, it is important that the mercury dose in the lamp being measured is close to the equilibrium location and vapor pressure. To achieve this goal follow LM-54. While T5 lamps are seasoned in the vertical orientation they are measured horizontally.⁷ All subsequent burning is done horizontally; making sure the cold spot is not moved above the horizontal plane.

6.2.3 Determination of Stabilization Time. Once pre-burning is complete and the lamps have been transferred to the test position in the photometric measurement system, the lamps shall be operated so as to attain stable light output. The time required to attain stabilization can vary considerably depending on whether the lamps are transferred from the pre-burn position hot (or cold), lamp wattage, whether the lamps are covered (or bare), and the lamp design (i.e., amalgam). Typical stabilization times for a range of lamp designs, with normal care used in handling the lamps, are 5 to 45 minutes (15 minutes is the most typical).

The following is a generally accepted four-step method for determining if a fluorescent lamp is stable:

Step 1. Take six measurements of the lamp light output at one-minute intervals (total time = five minutes).

Step 2. Determine the maximum measured value, the minimum measured value, and the average value of the six consecutive measurements. Calculate the percent stability by dividing the difference between the maximum and minimum values by the average value of the six consecutive measurements.

Step 3. If the value calculated in *Step 2* does not exceed 1.0 percent, the lamp is considered stable.

Step 4. If the value calculated in *Step 2* exceeds 1.0 percent the lamp is considered unstable and measurements in one-minute intervals have to be continued until the last six measurements meet the criteria in *Step 3*.

This method of determining lamp stabilization is strongly recommended. Variation in the acceptance criterion and the number and duration of the measurement interval provide different levels of lamp stability, either more or less severe. Should a different method or criteria be used it shall be so noted in the test report.

6.2.4 Lamp Circuit Switching. If a lamp is warmed up on one ballast and then switched to a different ballast or variable reactor for measurement, an additional period of operation in the measurement circuit is usually necessary to restore equilibrium. This extra time can be minimized if the two ballasts are electrically equivalent and if the lamp is switched without extinguishing its arc. If the lamp arc is extinguished, or if the warm-up ballast does not match the measurement ballast, additional operating time may be required to re-establish stabilization. When transferring lamps from a warm-up position to the photometric equipment, the pin connections and orientation used during warm-up shall be maintained. Careful lamp handling is particularly important during lamp transfer so mercury is not dislodged from the cool zones.

6.3 Use of the "Peak Lumen" Method

To reduce the test time, it may be desirable to take

measurements at peak light output. By careful correlation tests, it may be possible to relate peak to stabilized lumens by a constant, which will be unique to each lamp type. This correction factor will be equal to the stabilized lumens divided by the peak lumens. If lamps are read at peak and lumen data is presented in terms of the stabilized value, the peak measurement shall be multiplied by this correction factor.

Since a number of lamp design and process variations exist, considerable testing and experience with a given lamp type may be required to correlate peak and stabilized measurement results. Therefore, the stabilized method is preferred. If the peak method is used it shall be so noted in the test report.

6.4 Unusual Conditions

A lamp that shows visual temporal non-uniformities or other abnormal behavior is considered unstable. This can usually be detected with the naked eye; however, there are also instabilities that occur in the lamp which are not visible that may result in abnormal photometric and electrical measurements. If such instabilities are suspected to be present running a small permanent magnet along the length of the lamp may alleviate the instability. Unstable lamp voltage may also be due to such arc non-uniformities even when they are not visually observed. Turning the lamp off for approximately 15 seconds (without disturbing the lamp position) and then restarting it will usually eliminate lamp striations or swirling. The stabilizing procedures should then be resumed before any measurements are made.

6.5 Electrical Settings

The standard method of measuring a fluorescent lamp shall be with the lamp stabilized and operating on the reference circuit. The preferred method for taking measurements is with the lamp operating and stabilized at the specified input volts to the reference circuit. Alternatively, measurements may be taken with the lamp stabilized, at the rated lamp power, or at a specified current. The method used shall be indicated in the test report.

6.6 Electrical Instrumentation

6.6.1 Instrument Tolerance. Instrument manuals provide values for the tolerance (accuracy) of the instrument, normally as a percent of reading.^{8,9}

The instrument tolerance for the measurement of electrical characteristics shall be: voltage ± 0.5 percent or less, current ± 0.5 percent or less, and wattage ± 0.75 percent or less. (Refer to NIST 1297 for the conversion of instrument tolerance to measurement uncertainty.¹⁰)

Distorted waveforms are present in discharge lamp circuits. Therefore, all test instruments shall respond adequately to frequencies higher than the input line fundamental. The instrument(s) chosen for voltage and current measurements shall be of the true RMS type and shall have a specified tolerance and frequency response adequate to ensure that these requirements are met. For high frequency measurements, instruments shall have a minimum frequency response of 100 kHz.

6.6.2 Impedance Limitations. The voltage input of the multifunction meter shall have an input impedance greater than one megohm (1 M Ω) and the current inputs shall have impedances less than ten milliohms (10 m Ω).

7.0 Photometric Test Procedures

Lumen measurements may be done using *photo-detectors* or *spectral detectors*, with spectral detectors being preferred over photo-detectors.

7.1 Measurements with Photo-detectors

The photometer's photo-detector shall have a relative spectral responsivity, which approximates the $V(\lambda)$ function.¹¹ There are numerous sources of error in making photometric measurements.^{12,13} An assessment should be made of these errors and their impact on measurement results. Potential sources of error shall be identified and significant sources of error either corrected or reported. (See ANSI/IES LM-78-20 for details.¹⁴)

When lumen correction factors are used to correct measurement errors they shall be well documented and analyzed. If the documentation shows that the correction factor is at least partially based on spectral mismatch of the detector or on spectral non-linearity of the paint (inside an integrating sphere), the correction factor shall be lamp spectral power distribution specific.

Spectral mismatch errors and the methods for calculating and applying measurement corrections are of particular interest. The following discusses this error in detail:

Commercially available photo-detectors only approximate the standard $V(\lambda)$ response due to limitations in the manufacturing process. Generally, the closer the "fit" to $V(\lambda)$, the more expensive the detector. A numerical term used to express the accuracy of fit is f_1' . It is important to note that this term provides an indication of the average closeness of the detector response to the ideal curve (i.e., its overall conformance to the standard); f_1' does not provide information about the magnitude of deviation of the response from $V(\lambda)$ at specific wavelengths.¹³ To determine the magnitude of error of a particular measurement due to the photo-detector's spectral mismatch, the following shall be known:

- The relative spectral response of the detector
- The spectral distribution of the source being measured
- The spectral distribution of the source used to calibrate the detector

The relative spectral response of a photo-detector is often available from the manufacturer and it can be measured at the National Institute of Standards and Technology or at a commercial laboratory.** Provided that the necessary equipment and the expertise are available, the photo-detector response can be measured using reference standards. The spectral distributions of the source under test and the calibration source shall

** Contact data can be found on the IES website (available to members only): <http://iesmanufacturersdirectory.com/>. Just below the heading, click on Browse Categories. From the menu that appears, select Miscellaneous, and then Photometric Testing Labs. Directory users will need to contact laboratories directly to ascertain whether they can perform the desired test.

be determined using a spectroradiometer. A correction can then be calculated and applied to the measurement results.

7.2 Measurements with Spectral Detectors

Spectral lumen measurements¹⁵ have many advantages over photo-detector lumen measurements. There is no need for spectral mismatch correction due to non-flat sphere paint or the deviation of the photocell response from $V(\lambda)$, since every measured wavelength is individually calibrated. Also, while color measurements could be done with photo-detectors, CRI has to be determined spectrally providing all the information needed to calculate color and lumen values. A disadvantage could be a need of an absolute spectral scale (now available through the National Institute of Standard and Technology), which could be pricier to obtain than just an absolute lumen scale.

Instruments routinely used for spectral lumen measurements contain diode arrays or charge coupled devices (CCD). The CCD-based instruments provide faster measurements with little or no trade off in stability, reproducibility, and working range. While a modern photo-detector and amplifier can routinely provide linear response in a six order-of-magnitude range, diode arrays and CCDs have a practical limit of two to three orders of magnitude without hardware or software changes.

Since these instruments are based on single monochromators, the presence of stray light could affect the color measurements. If this becomes a problem, it can be corrected by using color corrections referenced to scanning double monochromator instruments or by stray light calibration of the spectroradiometers (now available through the National Institute of Standard and Technology). Also, the spectrum can simply be taken using scanning double monochromator instruments which tend to be slower than single monochromators.

7.3 Normal Intensity (Candela)

Directional light output may be measured with one or more detectors at an appropriate distance from the lamp. Such a device shall have a relative spectral responsivity which approximates the $V(\lambda)$ function. The recommended minimum distance is five times the lamp

length to keep the error caused by the assumption that the source is point will be less than 1.0 percent. Total luminous flux may be computed, if the flux/intensity ratio for the type of lamp is well established. The lamp shall be operated in a horizontal orientation unless the test requirements specify otherwise.

The intensity of lamps having a circular cross-section but a non-symmetrical distribution of light output around the lamp axis (e.g., reflectorized type fluorescent lamps) may be taken as the average of several readings made at points uniformly spaced around the lamp axis. At least four such readings shall be taken but more are preferable.

ANSI/IES LM-78-20 can provide information on sources of error associated with detectors, such as spectral mismatch.¹⁴

- Stray light
- Distance measurement
- Deviation from the inverse square law (angular size of source)
- Angular alignment of the lamp and detector

7.4 Intensity (Candela) Distribution

The distribution of luminous intensity around a lamp may be determined with a photometer similar to that used for normal luminous intensity measurements, provided there is a means for varying the angles between the detector and the lamp axis.

Potential sources of error, in addition to those in **Section 7.3**, which require consideration are:

- Error in the scan angles
- Spectral errors due to spectral reflectivity of turning mirror(s)
- Drafts induced by goniometer movement

Except for lamps with cold chambers or cold spots, the detector may be fixed and the lamp oriented at various angles in planes passing through the axis about which the lamp is moved. Readings are generally taken in ten-degree steps to avoid inaccuracy resulting from sudden changes in distribution.

Lamp types, which require that all excess mercury be condensed in pockets or cold chambers for proper stabilization, shall remain in a fixed horizontal orientation (or as recommended by the manufacturer). With fixed lamp orientation, a folded mirror goniophotometer can be used to direct the light to the detector, avoiding the need for an excessively large photometer while maintaining conformance with the inverse-square law. The readings are generally taken in ten-degree steps to avoid the inaccuracy that can result from sudden changes in light distribution.

7.5 Integrating Sphere Measurement

The integrating sphere method gives total luminous flux of the test lamp with one measurement.¹¹ Air movement is minimized and temperature within the sphere is not subject to the fluctuations usually present in a temperature-controlled room. If, however, the enclosure is small, accumulated heat from the lamp under test can raise the temperature within the sphere. To maintain an ambient temperature of $25\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F} \pm 1.8\text{ }^{\circ}\text{F}$), it may be necessary to keep the sphere open while the lamp is stabilizing, then close it just prior to taking the measurement. (Refer to ANSI/IES LM-78-20 for potential sources of error and methods of correction.¹⁴)

The spatial non-uniformity effect of the sphere response needs to be evaluated when measuring the lumen output of lamps having a circular cross-section but a non-symmetrical distribution of light output around the lamp axis (e.g., reflectorized fluorescent lamps).

7.6 Color Measurements

The measurement of color is specified in terms of the chromaticity coordinates and color rendering indices as defined by CIE^{3,11,12,15-17} and practical guidance for implementation in IESNA documents.^{15,16}

of testing, type of equipment and lamps. Typical items reported are:

- Date and testing agency
- Manufacturer's name and lamp designation
- Type of test
- Number of lamps tested
- Rated electrical values for the lamp tested
- Number of hours operated and operating cycle
- Ambient temperature and temperature correction factor, if any
- Lamp orientation during test
- Seasoning and stabilization times
- Type of standard used and correction factors, if any
- Photometric method
- Sphere diameter or test distance
- Circuit conditions and fixed parameters
- Measured light output (lumens and/or candelas) and electrical values (volts, amperes, watts) of each lamp and group averages
- Color data
- Spectral Power Distribution
- Flux/intensity ratio if lumens were calculated from intensity measurement data
- Equipment used
- Statement of uncertainty if appropriate
- Deviation from standard operating procedures
- Observations of any unusual behavior of lamps, e.g., range of measured lumen values when the stability reference limit is exceeded

8.0 Test Report

The test report shall list all significant data for each lamp tested, together with performance data. The report shall also list all pertinent data concerning conditions

INFORMATIVE REFERENCES

- 1 Illuminating Engineering Society. ANSI/IES LM-66-20, Approved Method: Electrical and Photometric Measurements of Single-Based Fluorescent Lamps. New York: IES; 2020.
- 2 [This number is not used.]
- 3 International Commission on Illumination (CIE). CIE 15.2-1986, Colorimetry. Vienna: CIE; 1986.
- 4 American National Standards Institute, American National Standard for Electric Lamps Single-Based Fluorescent Lamps — Dimensional and Electrical Characteristics, ANSI C78.901-2005. New York, NY: ANSI, 2005.
- 5 American National Standards Institute, American National Standard for Lamp Ballast — Line Frequency Fluorescent Lamp Ballast. ANSI C82.1-2004. New York, NY: ANSI, 2004.
- 6 Mertz, D. C. and Wainio, A. W., "New 800 and 1500 mA Fluorescent Lamps for Use as Lumen Standards," Illuminating Engineering, Volume 64, Number 11, November, 1969, Page 659.
- 7 IEC-60081 ED. 5.1 B:2002, Double-capped fluorescent lamps - Performance specifications.
- 8 ISO Guide to Expression of Uncertainty in Measurement, First Edition 1993, ISBN 92-67-10188-9.
- 9 Illuminating Engineering Society. ANSI/IES LM-28-20, Approved Method: Guide for the Selection, Care, and Use of Electrical Instruments in the Photometric Laboratory. New York: IES; 2020.
- 10 Taylor, B. N. and Kuyatt, C. E., Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results, NIST Technical Note 1297, 1994 Edition.
- 11 International Commission on Illumination (CIE). CIE 84-1989, The Measurement of Luminous Flux. Vienna: CIE; 1989.
- 12 International Commission on Illumination (CIE). CIE 53-1982, Methods of Characterizing the Performance of Radiometers and Photometers. Vienna: CIE; 1982.
- 13 International Commission on Illumination (CIE). CIE 69-1987, Methods of Characterizing Illuminance Meters and Luminance Meters. Vienna: CIE; 1987.
- 14 Illuminating Engineering Society. ANSI/IES LM-78-20, Approved Method: Total Flux Measurement of Lamps Using an Integrating Sphere Photometer. New York: IES; 2020.
- 15 Illuminating Engineering Society. ANSI/IES LM-58-20, Approved Method: Spectroradiometric Measurement Methods for Light Sources. New York: IES; 2020.
- 16 Illuminating Engineering Society. ANSI/IES LM-16-93, IESNA Practical Guide to Colorimetry of Light Sources. New York: IES; 1993.
- 17 International Commission on Illumination (CIE). CIE 13.3-1995, Method of Measuring and Specifying Color Rendering of Light Sources. Vienna: CIE; 1995.
- 18 Illuminating Engineering Society. ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020.

References are available from:

IES - Illuminating Engineering Society

120 Wall Street, Floor 17
New York, NY 10005-4001
Phone (212) 248-5000

ANSI - American National Standards Institute

Attn.: Customer Service
25 West 43rd Street
New York, NY 10036
Phone (212) 642-4900

CIE (Commission Internationale de l'Eclairage)

TLA-Lighting Consultants Inc.
7 Pond Street
Salem, MA 01970-4819
Phone (978) 745-6870

or in Canada:

National Research Council

c/o Alan Robertson
NRC Institute for National Measurement Standards
(NRC – INMS)
1200 Montreal Road
Ottawa, Ontario K1A 0R6, Canada
Phone (613) 993-9347

to perform photometric measurements made at the nominal (or other) current.

For fixed-current measurements, the setup is similar to that used for making a reference measurement, but the current to the lamp is controlled manually (with an electronic control circuit or control software). Use of the reference ballast specified in the lamp data sheet is not necessary, although it may be used in the measurement circuit as long as the input voltage can be adjusted to obtain the nominal current. Since the lamp current is controlled during this type of measurement, the value of the ballast impedance has minimal effect on the lamp's operation.

This measurement method helps eliminate several uncertainty components that would otherwise contribute significantly to reference measurements (for example; voltage measurement uncertainty, ballast impedance variations vs. temperature, and any effects from the parasitic capacitance found in wiring or other parts of the system). Thus results from fixed-current measurements can provide lower uncertainty measurements. Generally, performing both reference and fixed-current measurements on a given lamp should give similar results, as the lamp under reference conditions should operate at near the nominal current.

Informative Annex

Photometric Measurements of Lamps Using the Fixed Current Method

Note: The information in this Annex is not intended to be a specific recommended procedure, but is presented as reference information to more fully acquaint the user with some alternative methods of making photometric measurements on fluorescent lamps. Because of the wide range of potential applications for these lamps, there exists a strong interest in expanding or modifying the measurement procedure in order to obtain additional information on electrical and photometric characteristics.

Commercial electronic ballasts for use with linear fluorescent lamps are usually designed to keep the lamp current at a nominal level that is specified in the relevant lamp data sheet. Therefore, it may be necessary



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